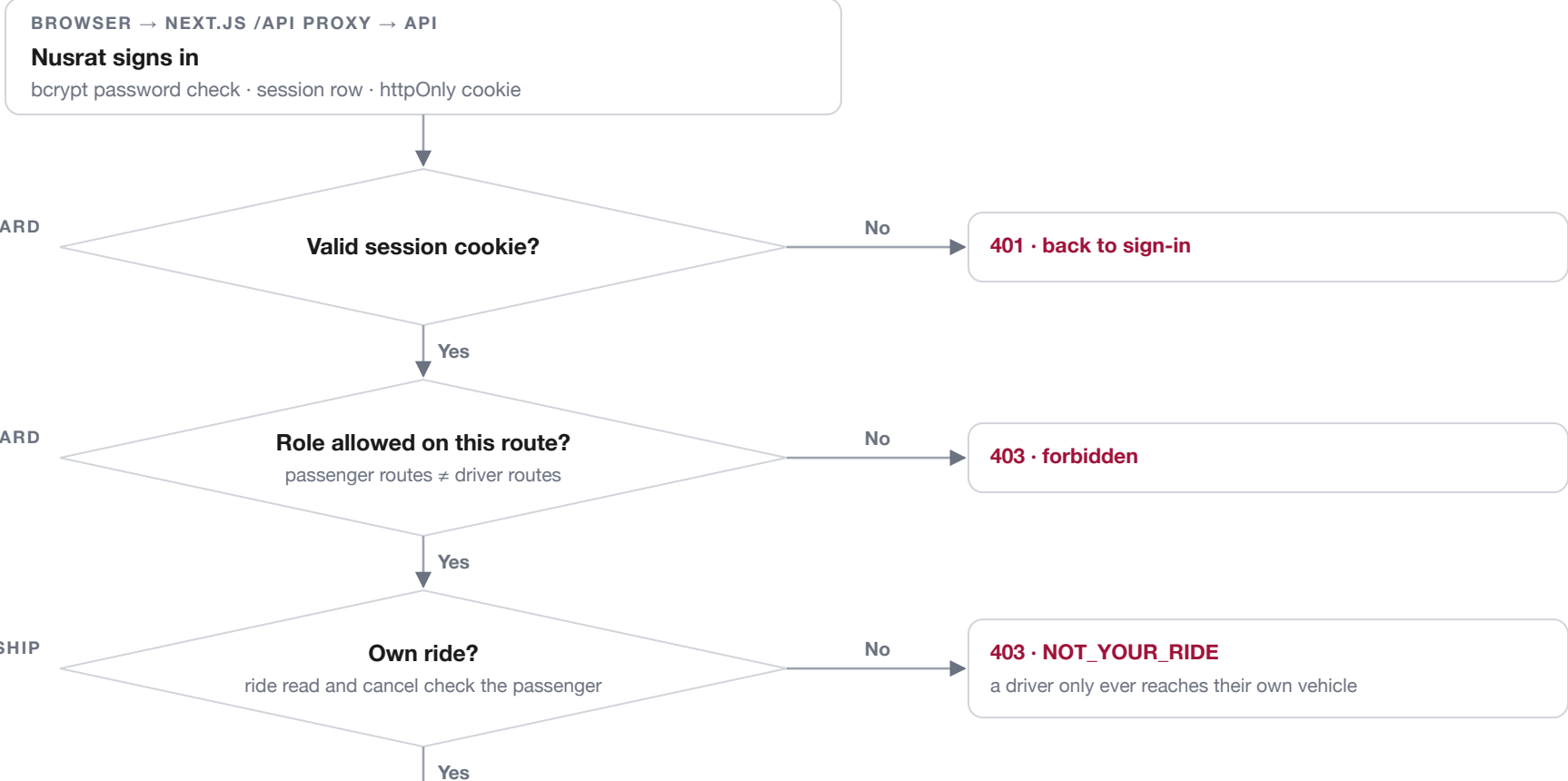


Dhaka Tesla Pool · Ride Flow

How one ride moves through the system · a Tesla picks people up along its route · side notes cover concurrent and edge cases

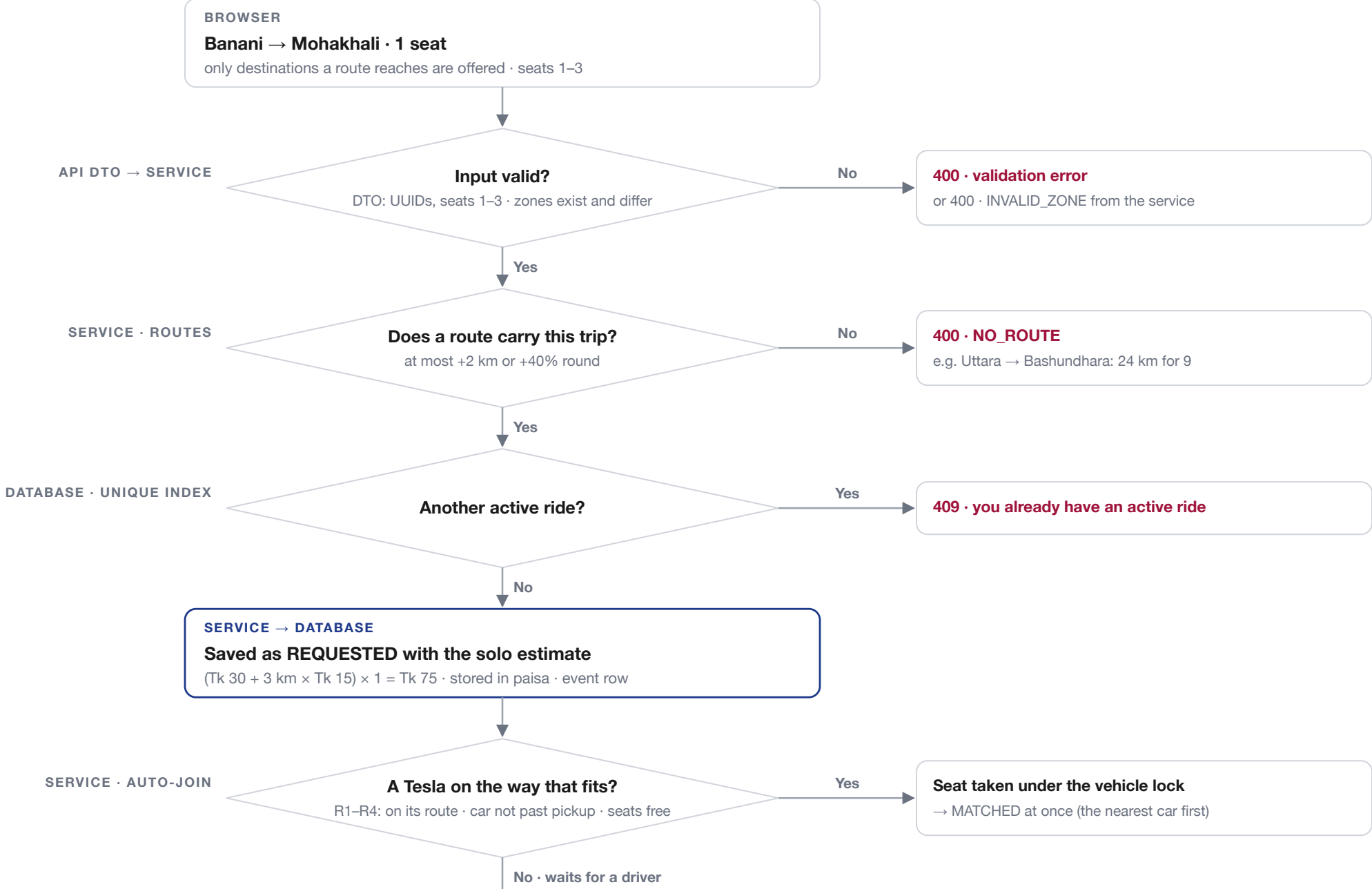
1 SIGN IN



Sign up

Passenger or driver, chosen first  
Driver: NID or passport, licence, car  
Unique email, phone, document, plate  
bcrypt · 5 attempts per minute

2 REQUEST A RIDE



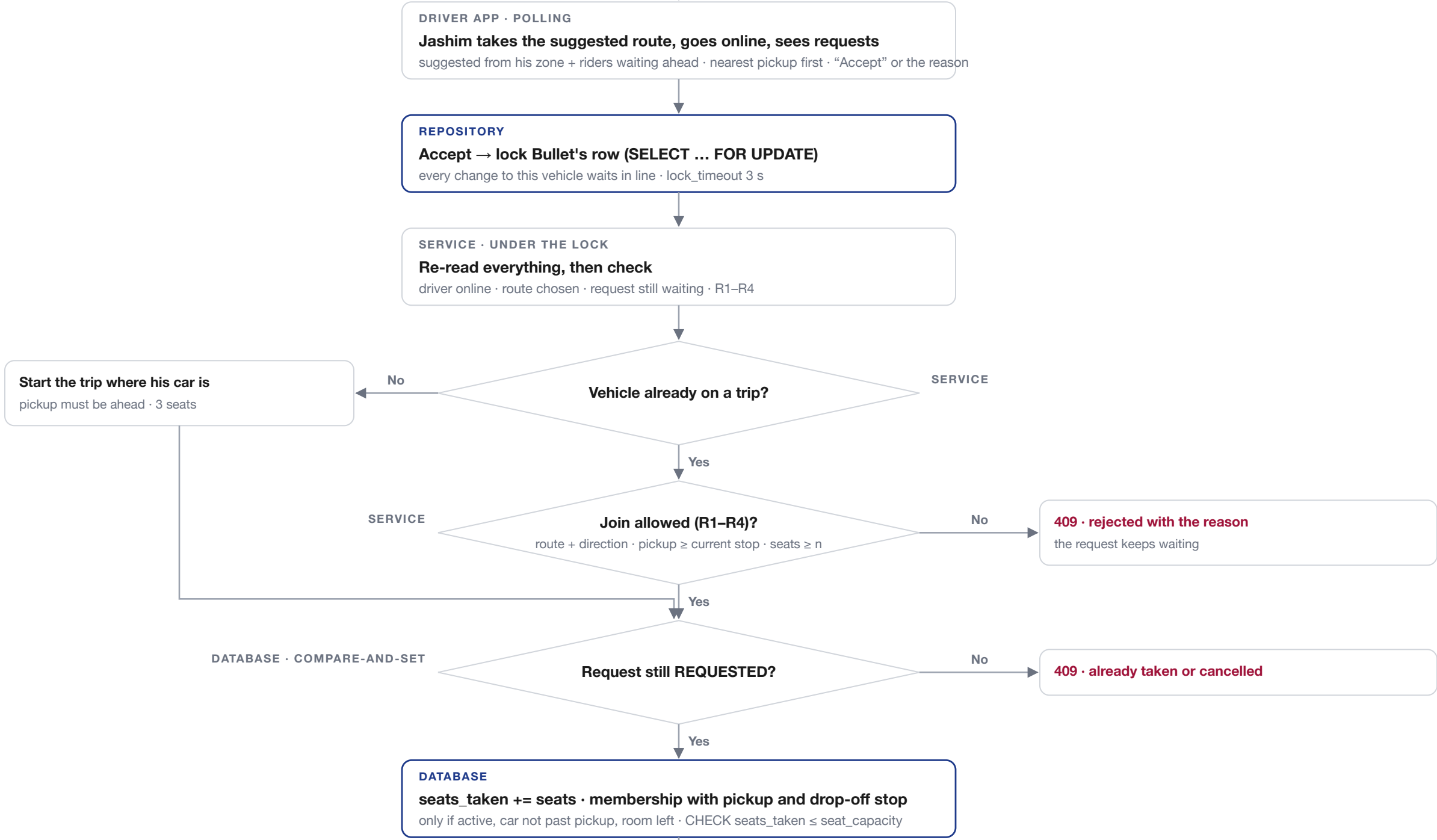
Routes (R1)

6 routes: 3 lines, both directions  
At most +2 km or +40% round  
Rafiq: 6 km for 4 km ✓  
Uttara → Bashundhara: 24 for 9 ✗

Who pays, who earns

Rider: (Tk 30 + km × Tk 15) × seats  
–20% if shared a hop · locked at drop-off  
Driver: Tk 10/km carried + Tk 20/pickup  
Platform keeps the rest (≥ Tk 10)

3 MATCHING



Last seat, many riders at once

Nusrat holds 2 of Bullet's 3 seats  
20 riders ask for the last one  
All wait for Bullet's lock  
Exactly 1 wins · 19 keep waiting

No trip yet, two accepts at once

First accept starts the trip  
Second waits, then joins it  
One trip, two passengers

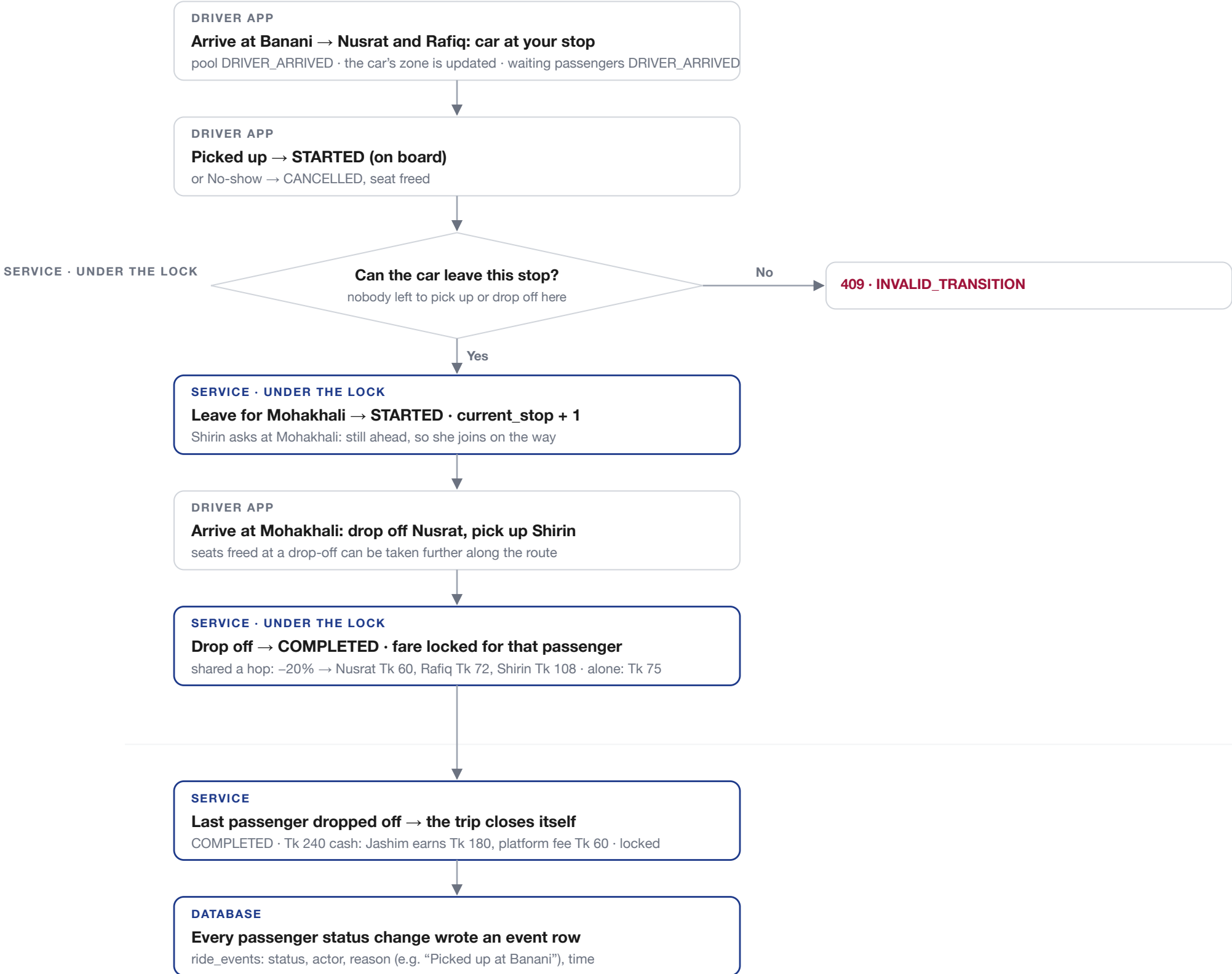
Retries and timeouts

Lock wait > 3 s → 503 BUSY  
Auto-join is best effort: if busy,  
the ride just keeps waiting

Database guarantees

seats\_taken ≤ seat\_capacity  
pickup\_stop < droptoff\_stop  
1 active pool per vehicle  
1 active membership per ride  
1 active ride per passenger

4 THE TRIP, STOP BY STOP



Route, location, online

System suggests, driver confirms  
Location follows the stops  
Route changes only between trips  
Offline refused while a pool is active

Leave vs join at the same stop

Car at Mohakhali, Shirin asks there  
Both wait for Bullet's lock  
Join first → the car waits for her  
Leave first → stop is behind: she waits

What the passenger sees

Polls her ride every 3 s (history 10 s)  
The route with the car on it  
Own status and own fare only  
Co-riders' first names only

Cancellation rules

Passenger: until picked up  
→ seats freed at once  
Driver at the stop: no-show  
Driver: before the first pickup →  
passengers back to REQUESTED  
Last one out → the trip closes